

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 1

CANDIDATE
NAME

ANSWERS

BIOLOGY
CLASS

REGISTRATION
NUMBER

BIOLOGY

Paper 2

8875/02

30 August 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Biology class, registration number and name on all the work you hand in.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in brackets [] at the end of each question or part question.

Section A

Write your answers in this booklet using dark blue or black pen.

Section B

Write your answers on the separate Answer Paper using dark blue or black pen.

For Examiner's Use	
Paper 1	
	/ 30
Paper 2	
Section A	
1	/ 10
2	/ 7
3	/ 12
4	/ 11
Section B	
5 / 6	/ 20
TOTAL	/ 90

This document consists of **15** printed pages.

[Turn over

Section A

Answer all the questions in this section.

- 1 Fig. 1.1 is a transmission electron micrograph of a mammalian cell.

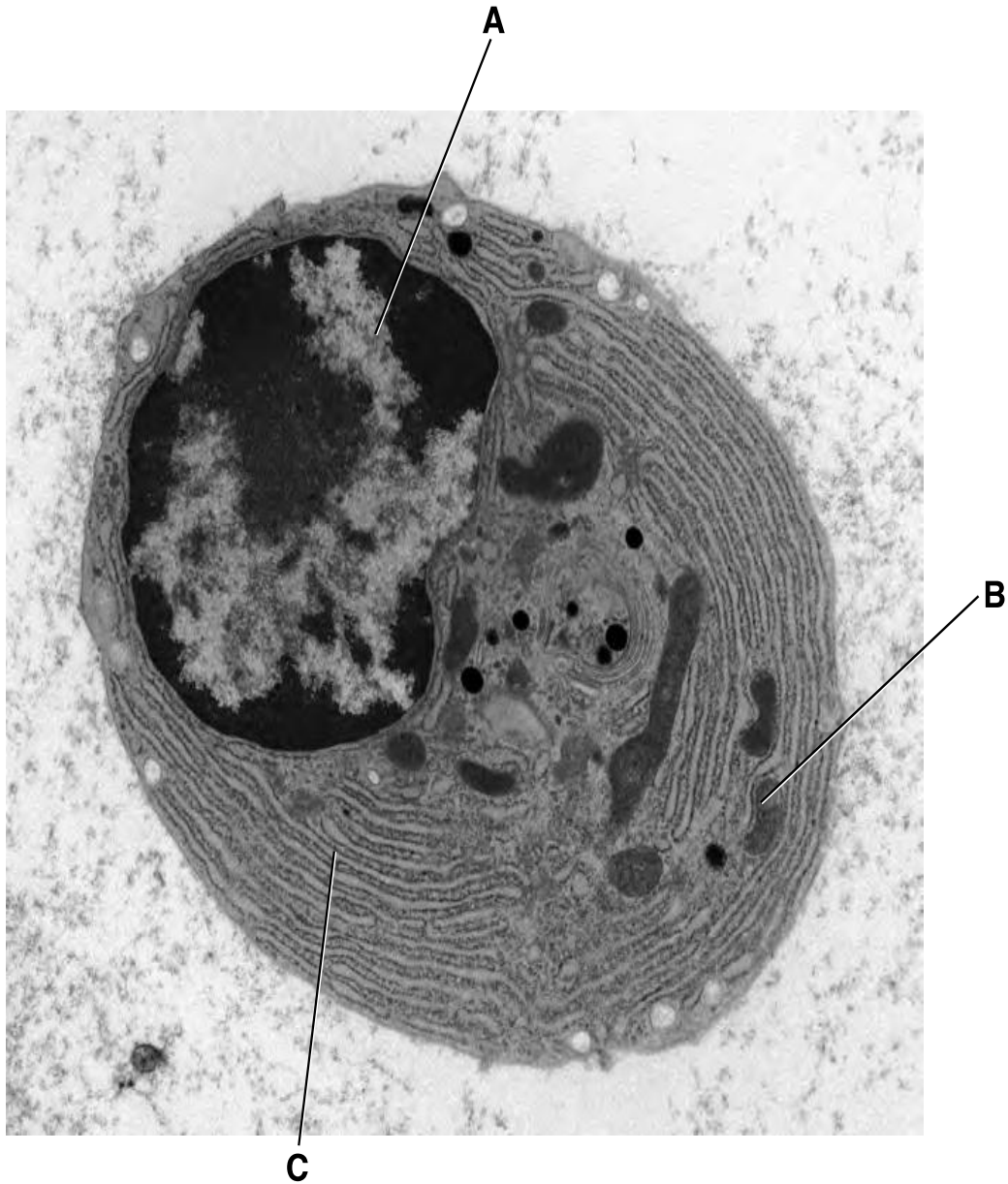


Fig. 1.1

- (a) Complete Table 1.1 to:

- name in full, structures **A**, **B**, and **C**.
- outline how each structure functions to contribute to its specific role in the mammalian cell.

Table 1.1

<i>structure</i>	<i>name of structure</i>	<i>function of structure within mammalian cell</i>
A	<i>A: nucleus / (eu)chromatin</i> <i>R: heterochromatin / chromosome</i>	• <i>Contain gene(s)/ genetic information/ genetic material / DNA that codes for proteins/ polypeptide</i> • <i>Site for transcription (occurring) leading to mRNA synthesis</i>
B	<i>mitochondrion</i>	• <i>Provides / synthesizes / produces / makes ATP for protein synthesis / exocytosis</i>
C	<i>Rough endoplasmic reticulum</i>	• <i>Synthesis / modification / processing / transport, of protein / polypeptide</i> <i>A: translation</i>

[3]

Table 1.2 below represents the ionic concentrations inside and outside a typical mammalian cell.

Table 1.2

	<i>normal conditions</i>	
	<i>concentration (mM)</i>	
<i>ions</i>	<i>intracellular</i>	<i>extracellular</i>
Na^+	5-15	145
H^+	7×10^{-5}	4×10^{-5}
Cl^-	5-15	110

(b) Explain how the intracellular concentration of Na^+ is maintained.

[1]

- *Sodium-potassium pumps pump out excess Na^+ → requires ATP for the active transport of this ion.*

Cholera, a disease caused by a bacterial infection of *Vibrio cholerae*, results in severe diarrhoea leading to dehydration. A toxin released by the bacterium causes the release of chloride ions (Cl^-) from cells lining the small intestines into the lumen and inhibits the uptake of sodium ions (Na^+) by these cells. Under normal circumstances, the Cl^- intracellular concentration is maintained due to the natural electrical negativity of the cell.

(c) By comparing normal and diarrhoeal situations, explain how this disruption of cellular ion concentrations would result in extreme dehydration. [4]

- *Under normal conditions, the chloride ions remain in the cells lining the lumen of the small intestine → creating an overall negative charge within the cells that drives positively charged sodium ions into the cells*
- *This creates a hyperosmotic condition in the cells/ lower water potential in cells (or content of the cell would contain more solutes than extracellular fluid) → Thus resulting in water moving into the cells by osmosis.*
- *When suffering from diarrhoea, there is increased concentration of Cl^- and Na^+ in the lumen of the small intestine.*
- *Elevated concentrations of these two ions results in the lumen of the small intestines becoming hyperosmotic/ having a lower water potential compared to the cells lining the intestines.*
- *This in turn causes large amounts of water through osmosis into the lumen of the intestine from the cells and blood stream → Water passes out of the body as diarrhoea, leaving the body dehydrated.*

(d) One of the antibiotics used in the treatment of cholera is tetracycline. Tetracycline causes the death of the pathogen by binding to ribosomes and inhibiting protein synthesis.

(i) Suggest one ways in which tetracycline acts at ribosomes to inhibit protein synthesis. [1]

- *Binding of tRNA is prevented → no codon-anticodon binding and peptide bond formation is prevented*
- *mRNA attachment is prevented → ribosome movement along the mRNA is hindered / prevented*
- *Inhibits the association of large and small subunits → ribosome movement along the mRNA is hindered / prevented*

(ii) Tetracycline does not harm mammalian cells. Suggest an explanation for this. [1]

- *Cell surface membrane of mammalian cell is impermeable to tetracycline*
- *Possibility of tetracycline to be broken down by enzymes / be degraded before entry into the cell*
- *Tetracycline targets prokaryotic ribosome and the structure of eukaryotic ribosome is different → difference in ribosomal structure*

[Total: 10]

- 2 (a) Pituitary dwarfism is a sex-linked inherited condition in humans in which affected individuals have very short limbs. The family tree, in Fig. 2.1, shows part of one affected family.

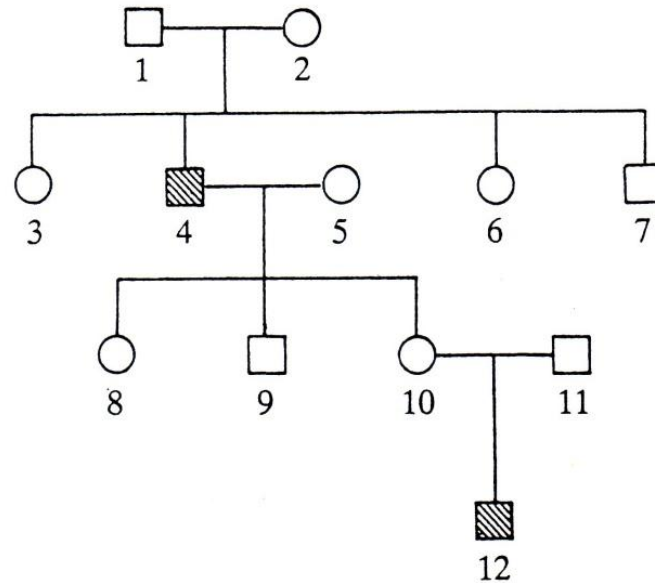


Fig. 2.1

- (i) Explain what is meant by 'sex-linked inherited condition'. [1]

- *A character that is controlled by a gene situated on the X-chromosome only*

- (ii) With reference to Fig 2.1, explain the mode of inheritance using two pieces of evidences. [4]

- *If the allele for dwarfism is dominant, the affected male, 4, who has only one X chromosome should pass this allele to all his daughters, who should then be affected.*
- *However, his daughters, 8 and 10 are normal. Hence allele for this condition is not dominant but recessive. OR*
- *Male 4 receives its only X chromosome carrying the allele for dwarfism from his mother 2*
- *Hence his mother 2 is a heterozygote with an allele for normal limb and an allele for dwarfism*
- *Since she is normal, the allele for dwarfism must be recessive to the allele for normal limb.*

A series of breeding experiments were carried out with a certain species of mice. A cross between pure breeding males with red eyes and bent tails and females with black eyes and straight tails produced offspring all with red eyes and bent tails. The same results were obtained when the parents of the same pure –breeding phenotypes were used again but with the sexes reverse.

(b) Suggest a reason for carrying out the cross with the sexes reversed. [2]

- *To check whether the genes are sex-linked.*
- *If sex-linkage, then offspring phenotypic ratio will be different*

[Total: 7]

3 Fig. 3.1 illustrates what happens on thylakoid membrane during daytime.

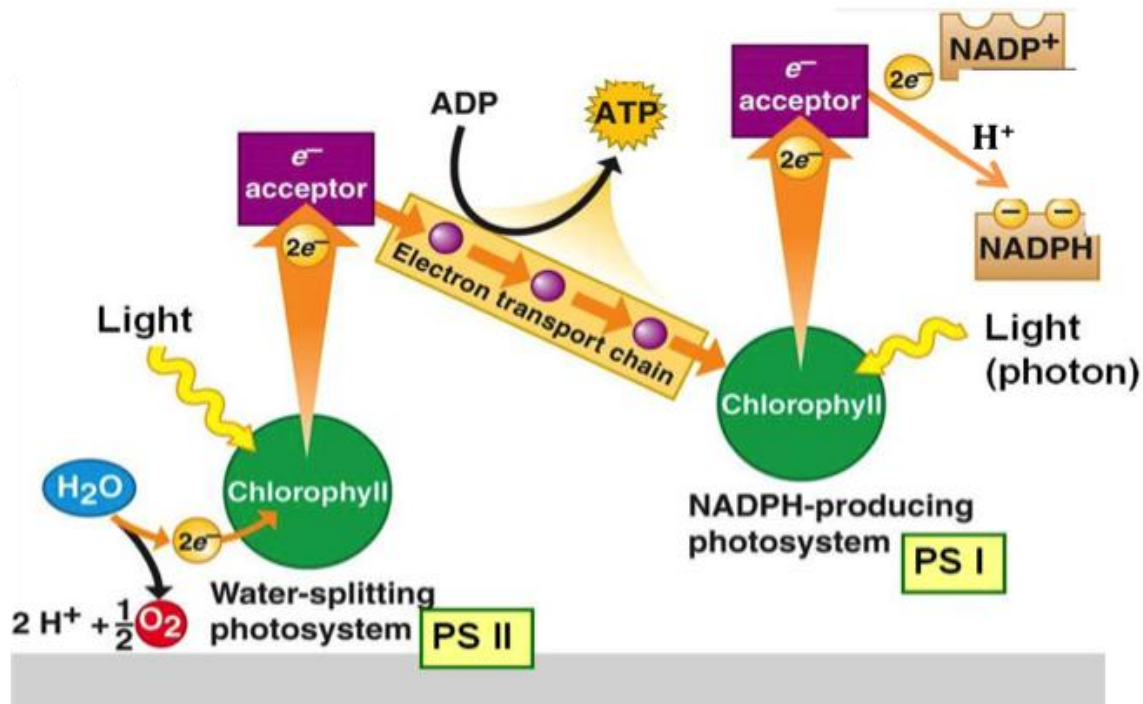


Fig. 3.1

(a)

(i) Describe how Electron Transport Chain leads to ATP production, stating clearly the location when ATP is synthesized. [3]

- When electrons are transferred through a series of electron carriers arranged in a progressively lower energy fashion, energy is released during each transfer
- The energy is used to actively transport H^+ against its concentration gradient from stroma to the thylakoid space.
- Chemiosmosis occurs when the H^+ flows down its concentration gradient via facilitated diffusion through ATP synthase, driving the synthesis of ATP from ADP and P_i in the stroma.

(ii) Compare the initial electron donor and final electron acceptor between oxidative phosphorylation and photophosphorylation. [2]

	<i>Oxidative phosphorylation</i>	<i>Photophosphorylation</i>
<i>Initial electron donor</i>	<i>NADH and FADH₂</i>	<i>H₂O</i>
<i>Final electron acceptor</i>	<i>O₂</i>	<i>NADP⁺</i>

(b) A point mutation results in dysfunctional NADP^+ reductase. State and explain how the concentration of Ribulose 1,5 – Bisphosphate in the chloroplasts would be affected. [3]

- *NADP^+ cannot be reduced to NADPH which is required as a reducing power for the conversion of 1,3-bisphosphoglycerate to glyceraldehydes 3-phosphate.*
- *The subsequent regeneration of Ribulose 1,5 – Bisphosphate is prevented while carbon fixation continues to occur,*
- *depleting the supply of Ribulose 1,5 – Bisphosphate, causing its concentration to drop.*

(c) Glyceraldehyde 3-phosphate exits the Calvin cycle for production of glucose, which will be used for cell wall formation. Explain how the structure of cellulose allows it to serve its function. [4]

- *About 10,000 β -glucose units with alternating units rotated 180° in space are linked by $\beta(1 \rightarrow 4)$ glycosidic bonds to form long and linear chains.*
- *Large number of H-bonds / cross-links is formed between the linear cellulose chains due to the large number of β -glucose units with protruding $-\text{OH}$ groups in each chain.*
- *These cross-linked chains are associated to form microfibrils, which are then meshed into a macrofibril matrix .*
- *The resultant great tensile strength & insolubility in water allows cellulose to be the main component of plant cell wall, providing structural support to plant cells. The porous matrix allows free passage of water molecules.*

[Total: 12]

- 4 The corn borer, *Ostrinia nubilalis*, is an insect pest of maize. The larvae are caterpillars that eat the leaves of the maize plants. The adults can fly. Adult corn borers do not feed on maize plants.

Much of the maize that is grown in the USA has been genetically modified to produce *Bt* toxin, which is lethal to insects that feed on the leaves. However, many populations of the corn borer have now evolved resistance to the *Bt* toxin.

- (a) Explain how this resistance could have evolved. [3]

- *Genetic mutation / random changes in corn borer*
- *Caterpillars / corn borers, with mutation, more likely to survive and have selective advantage*
- *(Adults with this mutation) likely to breed*
- *Mutated genes / resistance alleles will be passed on to next generation*
- *Increase in frequency of allele for resistance*

Fig. 4.1 shows the male and female flowers of maize.

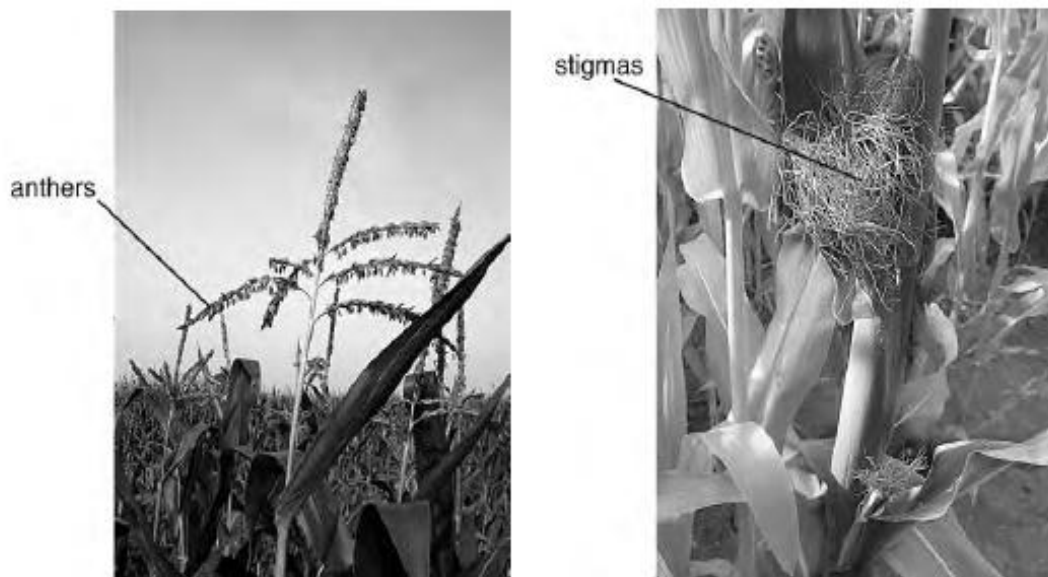


Fig. 4.1

Cultivated maize is related to a wild Mexican grass, teosinte. Teosinte looks very different from maize. In particular, teosinte has a very hard layer surrounding the fruit, making it impossible to use as an edible grain.

Archaeological excavations have found that maize in its edible form dates back at least 4000 years. It has been argued that the structural differences between teosinte and maize grains are so great that it is unlikely that maize could have been bred from teosinte so long ago.

Investigations have been carried out into the genetic differences between teosinte and maize. A gene on chromosome 4, *tga 1*, containing 1042 base pairs, was found to be responsible for all of the structural differences between teosinte and maize.

Fig. 4.2 shows the DNA base sequence of the only part of *tga 1* that always differs between teosinte and maize.

teosinte	GAT	TGG	GAT	CTC	AAG	GCG	GCG	GGC	GCG	TGG
maize	GAT	TGG	GAT	CTC	AAC	GCG	GCG	GGC	GCG	TGG

Fig. 4.2

(b) Outline the principles of electrophoresis as used in sequencing this DNA. [3]

- *Cutting of DNA into restriction fragments by the use of restriction enzymes*
- *DNA sample is loaded onto agarose gel and electric current is applied*
- *Fragments will migrate towards the anode (positively charged electrode) with shorter fragments migrate/ travel faster than longer ones*
- *Visualization of DNA fragments through the use of ethidium bromide to intercalate the DNA and UV light*

(c) Suggest how the difference in the base sequence of the *yga 1* gene shown in Fig. 4.2 could cause large differences in phenotype between teosinite and maize. [2]

- *Changes have been made to primary structure / secondary structure / tertiary structure / folding / 3D conformation*
- *Protein / enzyme cannot carry out its function*
- *Product could be an enzyme that is essential for a metabolic pathway*
- *Product could control the expression of another gene / series of genes*

(d) Account for the role of **three** types of RNA in producing the protein expressed from yga 1 gene. [3]

- *Sequence of bases in mRNA / ref to codons determine sequence of amino acids*
- *Different tRNA for each amino acid / tRNAs bring the correct amino acids to the ribosome for translation*
- *mRNA codon determines which tRNA binds (to ribosome), ref to codon-anticodon pairing / complementary base pairing*
- *rRNA is a structural component of ribosomes, which are the sites of protein synthesis*

[Total: 11]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 **(a)** With reference to specific genes, discuss how point mutations and chromosomal mutations can lead to cancer. [12]

- *Proto-oncogenes such as Ras gene codes for Ras protein which is involved in promoting mitosis in the presence of signaling molecules such as epidermal growth factor.*
- *Gain-of-function mutations change the proto-oncogene to oncogene, causing production of hyperactive Ras protein or over-production of Ras protein.*
- *Base pair substitutions in the coding sequence can results in missense mutations, coding different amino acids and resulting in altered three dimensional conformation of the protein. For instance, the GTPase active site could be altered and is no longer able to catalyse hydrolysis of GTP bound to Ras protein, making it perpetually active / hyperactive.*
- *Base pair substitutions in the regulatory sequences such as promoter and enhancer of the Ras gene can transform the promoter into a more powerful regulator of gene transcription, resulting in large quantity of Ras protein synthesized.*
- *Chromosomal translocation can cause Ras gene next to a stronger promoter, resulting in over-production of Ras protein.*
- *Aneuploidy as a result of non-disjunction during mitosis can lead to more than 2 copies of the Ras gene present in a cell, resulting in over-production of Ras protein also.*
- *Tumor suppressor genes such as p53 gene codes for p53 protein which is a transcription factor switching on genes whose products are involved in repairing DNA damage or triggering apoptosis should the damage is beyond repair.*

- *Point mutations and chromosomal mutations can lead to loss-of-function mutation in p53 gene, leading to nonfunctional p53 protein or inadequate amount of p53 protein.*
- *Base pair substitutions in the coding sequence of p53 gene can lead to missense mutations, coding for a different amino acid. The tertiary structure of p53 protein is altered, rendering it dysfunctional.*
- *Base pair substitution in the regulatory sequence such as promoter or enhancer of p53 gene can lead to a non-functional promoter or enhancer not recognizable by transcription factors and RNA polymerase, hence leading to transcriptional inactivation. No or insufficient p53 protein is produced.*
- *A chromosomal translocation of p53 gene next to a non-promoter region such as centromere or other repeated sequence cause the gene to be silent / no expressed.*
- *Aneuploidy as a result of non-disjunction during mitosis can lead to only 1 copy of p53 gene present, increasing the likelihood of losing functional p53 proteins as a mutant allele will not be masked by the other normal allele.*
- *When both p53 alleles undergo loss-of-function mutation, no functional p53 proteins are produced to arrest cell cycle when there is severe DNA damage, causing genetic instability.*
- *When mutations accumulate in the genome, especially mutations in other tumor suppressor genes and proto-oncogenes occur, cells will undergo mitosis uncontrollably, leading to cancer.*

- (b) Describe homologous chromosomes and distinguish their behaviors between mitosis and the first division of meiosis. [8]

Similarities:

- Same number of genes;
- Carry the same genes / carry code for the same characteristics;
- Same relative position / loci along the chromosome;
- Same length/ size/ shape;
- Position of the centromere is the same;

Differences:

- Different alleles/ carry genes of an alternative kind;
- Each originates from different parents;
- Different sequence of bases / nucleotides

STAGE	MITOSIS	FIRST DIVISION OF MEIOSIS
Prophase	No synapsis / no pairing of homologous chromosomes No crossing over / no chiasma formation	Synapsis occurs / homologous chromosomes pair up to form bivalents Crossing over occurs between non-sister chromatids of homologous pair at chiasmata.
Metaphase	Chromosomes arrange lengthwise / on the same plane / in a single row on the equator of spindle. No independent assortment since sister chromatids are identical.	Homologous chromosomes lie side-by-side / equidistant on either side of the equator. Independent assortment of alleles occur.
Anaphase	Centromeres divide Sister chromatids separate OR Separating chromatids are identical	Centromeres do not divide Homologous chromosomes separate OR Separating chromosomes are not identical.

- 6 (a) Outline the procedure for cloning an eukaryotic gene in a bacterial pUC19 plasmid using *Escherichia coli* cells to produce eukaryotic proteins. Your answer should also include the method used to avoid intron associated problem, the transformation of bacteria and the identification of the desired bacteria colonies. [12]

Synthesis of cDNA (max: 3 marks)

- Starting with total mRNA extracted from the cells examined, reverse transcriptase is used *in vitro* to make single-stranded DNA transcripts of the mRNA.
- Following enzymatic degradation of the mRNA, a second DNA strand, complementary to the first, is synthesized by DNA polymerase.
- This double-stranded DNA, called complementary DNA (cDNA), is then modified by the addition of restriction enzyme recognition sequences at each end.

Preparation of recombinant plasmid (max: 3 marks)

- Cut the plasmid and cDNA using the same restriction enzyme to generate complementary sticky ends
- The sticky ends of the restriction fragment can form hydrogen bonds (temporary associations) with complementary sticky ends of plasmid that is produced by the same restriction enzyme
- DNA ligase will now be used to join the 2 DNA fragments (plasmid DNA and cDNA) together by catalysing the formation of phosphodiester bonds between adjacent nucleotides

Transformation of bacteria (max: 2 marks)

- Bacterial cells can be artificially transformed. Chemical competence is conferred to bacterial cells by re-suspension in CaCl_2 solution at 0°C .
- Under these conditions, the Ca^{2+} ion is thought to create pores in the membrane, assist binding of the DNA to the cell membrane and mask the negative charge on the DNA, easing its passage through the hydrophobic cell membrane.
- The DNA is forced into the cells by applying a short 42°C heat shock, which results in a thermal current that sweeps the DNA into the cells.

Identification of desired colonies – works on the concept of insertional inactivation (max: 4 marks)

- Culture the bacterial cells in a medium containing ampicillin, X-gal and IPTG
- When bacteria containing non-recombinant plasmid are grown in the presence of IPTG (an artificial inducer of the lac operon), the product of lacI gene (i.e. the repressor)

cannot bind to the operator of the *lacZ'* gene. The *lacZ'* gene is thus transcribed and the resulting mRNA is translated to functional β -galactosidase.

- In the case when a gene of interest is inserted at one of plasmid's unique cloning sites, the coding sequence of β -galactosidase will be disrupted by insertional inactivation.
- Consequently, only bacteria containing non-recombinant plasmid can produce functional β -galactosidase that hydrolyses X-Gal to a blue product, which makes the colony appear blue. Bacterial colonies containing recombinant plasmid will appear white since they lack functional β -galactosidase.

(b) Discuss the sources of variation important for natural selection to occur. [8]

- In order for favorable genes to be selected, there must be genetic variation in a population.
- Such genetic variation might be inherited if mutation took place in the formation of gametes.
- Sources of such variation are as follows:
 - (A) Crossing over between 2 non-sister chromatids of homologous chromosomes during prophase I of meiosis will produce new linkage groups.
 - (B) The random alignment and independent assortment / segregation of bivalents / homologous chromosomes during metaphase I and anaphase I of meiosis respectively, as well as the random alignment and separation of chromatids during metaphase II and anaphase II respectively, will result in gametes with different chromosome combinations.
 - (C) Random mating between males and females in a population will lead to further genetic variation among the offspring, as different parents have different sets of genes to pass on to their offspring.
 - (D) Random fusion of a male gamete and a female gamete to form a diploid zygote will also increase genetic variation among the offspring, as each parent can produce a variety of gametes, and only one of them will be used to form the zygote.
 - (E) Gene mutations such as:
 1. Substitution of one nucleotide in the DNA nucleotide sequence by another nucleotide with a different base
 2. Deletion of one or more nucleotides from the DNA nucleotide sequence
 3. Insertion of one or more nucleotides into the DNA nucleotide sequence
 4. Duplication of a section of the DNA nucleotide sequence
 5. Inversion of a section of the DNA nucleotide sequence

may code for an incorrect amino acid sequence, which may produce a malfunctioning protein.

(F) Chromosome mutations / aberrations such as:

- (i) Translocation of part of the chromosome to another non-homologous chromosome
 - (ii) Duplication of part of the chromosome
 - (iii) Inversion of part of the chromosome
 - (iv) Deletion of part of the chromosome
- will change the structure of the chromosome.

(G) Chromosome mutations / aberrations such as:

- (i) Aneuploidy (a condition in which one or more chromosomes are either missing or present in more than the typical number of copies)
- (ii) Polyploidy (a condition in which there are more than two complete sets of chromosomes)

will change the total number of chromosomes.

- Aneuploidy is caused by non-disjunction (failure of homologous chromosomes / sister chromatids to separate to opposite poles during meiosis).
- With unequal numbers of chromosomes in the gametes, individuals with abnormal number of autosomes (e.g. Down's syndrome) or sex chromosomes (e.g. Turner's syndrome) can potentially result.
- Should all pairs of homologous chromosomes fail to separate to opposite poles during meiosis, diploid gametes will be produced.
- Polyploidy may result when a normal haploid gamete fuses with a diploid gamete to form a triploid organism, or when two diploid gametes fuse together to form a tetraploid organism.

(H) Phenotypic variation between organisms with identical genotypes may arise from environmental effects.